Participant Guide, Day 2: Build the analyst

Tuesday 3 November 2026 | 08:30 to 15:00 | Fictional airport

Today's goal

By 15:00 you have built an agent that uses a database tool and a document tool, you have measured and tuned it, and you have tried to break it.

Agenda

Time	Block
08:30	Quiz and SQL warm-up
09:00	Lab 4: RAG, properly (notebook 04)
10:30	Break
10:45	Lab 5: Build the agent (notebook 05)
12:45	Lunch
13:30	Lab 6: Tune, test, attack (notebook 06)
14:50	Wrap-up

Lab checklists

Lab 4, notebook 04 (90 min)

- ☐ Baseline hit@1 and hit@3 written down
- ☐ Compared chunking and embedding sizes
- ☐ Turned on the status filter and saw the draft question change
- ☐ Added keyword search and saw which questions improved
- ☐ Added three golden questions of my own

Lab 5, notebook 05 (120 min)

- ☐ Tested the SQL tool with a good query and a bad query (DROP)
- ☐ Tested the document tool, with and without the draft flag
- ☐ Read the nine rules in the system prompt
- ☐ Ran the five beats and read the tool log after each
- ☐ Verified at least three numbers with my own SQL

Lab 6, notebook 06 (80 min)

- ☐ Compared low, medium and high thinking levels on four questions
- ☐ Saw the effect of a tiny output cap
- ☐ Ran five safety attacks and recorded each result
- ☐ Estimated monthly cost for 25 users

The five beats

1. Is the feeling real? Which weekday has the longest wait at Terminal 1?
2. When, and why? Hour by hour, waits and counters.
3. What does it cost? Lost spend, missed flights, estimated over the year.
4. What do the rules say? Maximum wait and minimum counters, approved documents only.
5. The aha: which clauses are we breaching, what is the cost, what must we do next?

What a strong answer looks like

- Wait and share over 30 minutes against the 15-minute and 5-percent limits
- Counters open against the number required
- Cost scaled by 25, described as an upper bound
- Clause numbers and the expired exception
- The next step required by the escalation procedure

Red flags: a number with no query behind it; 12 or 20 minutes presented as the current limit; averages multiplied by 25; the wrong hour column; raw feedback instead of the clean view.

Reading the tool log

Which tools ran? Which SQL? Which sections came back, and were they approved? How many turns?

Glossary

Term	Meaning
Agent	Model plus tools plus a loop
Tool	A function the model may ask your code to run
Harness	The guardrails around the model: read-only, row cap, logs, filters
Golden set	Questions with known answers, used to test every change
Hit at 1 or 3	The right section was first, or in the top three
Hybrid search	Meaning search plus keyword search

Term	Meaning
Thinking level	How much hidden reasoning the model does

Homework (optional)

Notebook 06 sections 7 and 10. Think about which of the seven capstone scenarios you would like to take tomorrow.